

# NIKHIL KUMAR

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## TECHNICAL SKILLS

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- Machine Learning, Deep Learning
- Programming : Python, C++, Data Structures & Algorithms
- Data Analytics : Python (NumPy, Pandas, Matplotlib, Seaborn), SQL, Excel, Power BI

## SOFT SKILLS

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- Problem-Solving , Communication Skills , Team collaboration

## EXPERIENCE

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**Design Practicum | IIT MANDI | Feb 2023 - May 2023**

**Hand Gestures to Speech Converter gloves for Dumb people**

- Technologies Used: Arduino IDE, MPU6050 gyroscope module, flex sensors, nRF24L01 module.
- Captures hand movements using MPU6050 and flex sensors.
- Processes and compares gesture data with pre-recorded data stored on an SD card.
- Training Mode: Stores gesture data and associates it with audio files.
- Testing Mode: Performs real-time gesture recognition using absolute mean error comparison.
- Output: Converts recognized gestures into speech by transmitting corresponding audio files to a speaker via the nRF24L01 module.

## PROJECTS

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**Automated Semantic Image Segmentation for Autonomous Vehicles | Jan 2026**

- Engineered and trained a symmetrical U-Net Convolutional Neural Network (CNN) from scratch in TensorFlow/Keras to handle pixel-level semantic image segmentation.
- Implemented an encoder-decoder architecture leveraging Skip Connections to concatenate localized spatial features, directly preventing loss of boundary features.
- Processed and pipelined self-driving vehicle scene data from the CARLA Simulator using efficient batching, caching, and custom matrix normalization techniques.
- Utilized Sparse Categorical Crossentropy loss and the Adam Optimizer to map distinct multiclass segmentations for real-time roadway environmental tracking.
- Validated predictions by plotting categorical precision metrics and generating visual validation masks to verify real-time mapping performance.

**binary image classifier utilizing Transfer Learning and the MobileNetV2 CNN architecture | Feb 2026**

- Developed a computationally efficient binary image classifier utilizing Transfer Learning , achieving an overall validation accuracy of 96.92 %.
- Engineered a robust image preprocessing and data augmentation pipeline using the TensorFlow/Keras Sequential API, incorporating random flips and rotations to artificially expand the dataset and improve model generalization.
- Customized a pre-trained ImageNet base model by designing a new classification head with Global Average Pooling and a 20% Dropout layer, effectively mitigating overfitting risks.
- Implemented a two-phase training protocol: freezing the base network for initial training, followed by fine-tuning the top 35 layers using a BinaryCrossentropy loss function and a decayed Adam optimizer learning rate to capture dataset-specific features.

## EDUCATION

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**B.Tech. Mechanical Engineering | Indian Institute of Technology, Mandi**

Aug 2021 - May 2025

**Subhash Smarak Inter College, Saket Nagar , Kanpur Nagar**

Class XII - 80 % April 2021 , UP Board

**Shivaji Inter College, Keshav Nagar, Kanpur Nagar**

Class X - 87 % April 2019, UP Board

## CERTIFICATIONS

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- **Convolutional Neural Networks** : DeepLearning.AI, Jan 2026
- **Neural Networks and Deep Learning** : DeepLearning.AI, Dec 2025
- **Structuring Machine Learning Projects** : DeepLearning.AI, Dec 2025
- **Database and SQL** : Infosys | Springboard, Jan 2025